

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 415 — Past Questions & Materials
by Renance

Course: MEE 415
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MEE 415 Turbomachinery

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING q-- (so

FIRST SEMESTER EXAMINATION 2024/2025 SESSION 7

COURSE CODE AND TITLE: MEE 415 - TURBOMACHINERY Sha

TIME ALLOTTED: TWO HOURS, 30 MINUTES (2HRS: 30 MINUTES)

INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER 3 .

Pe ty Oe

QUESTION 1 whe Q

(a) A Pelton wheel is supplied with water under a head of 35 mat the rate of 40.5 kilo litres/min. The an bucket speed is 13m/s. Calculate

bucket deflects the jet through an angle of 160° and the me!

power and hydraulic efficiency of the Turbine wie eels

(b) The shaft of a small turbine turns at 20000 RPM and the blade speed is $U = 250$ m/s. The axial velocity leaving the stator $V_2 = 175$ m/s, The angle at which the absolute velocity leaves the stator is $\alpha_2 = 67^\circ$, the relative velocity leaves the rotor at $\beta = -60^\circ$ and the absolute velocity leaves the rotor at $\alpha_3 = -20^\circ$. Find the mean radius of the blade, fo, V5.3, V2 the Specific power delivered by the stage and the degree of reaction. — ~ ve ; ea RI

z hh - we r= yen ~T # | [25 Marks]

QUESTION 2 tar.

(a) Show that the maximum blade efficiency with equiangu equal to the maximum utilization for impulse turbine

(b) Develop the expression for Degree of Reaction as a function of Kinetic Energy and the Static component

uw

lar blade for impulse turbine is

{i5 Marks]

QUESTION 3

(a) Develop the combined velocity triangle for a Curtis Turbine

(b} Derive the expression for the blade efficiency and work done in the lasi row of a Curtis Turbine

(Note: all symbols used in the developmen and derivations should be clearly defined)

(15 Marks]

QUESTION 4

(a) Draw the velocity triangles for the different conditions of Francis Turbine

(b) The performance of a turbomachine has 'seen expressed as a function of density of the

\ fluid, ρ , speed of the rotor, N , Characteristics diameter, D , Discharge, Q , Gravity headm"
 gH , power developed, P , and fluid viscosity, μ . Using Buckingham Pi theorem, develop the dimensionless numbers

(15 Marks]

QUESTION 5

(a) Give complete classification of fluid machines = 7 ae = ficd ter

— (b) Compare positive displacement machines and Turbo machines... tape te

- (c) An output of 10 kW was recorded in a turbine, 0.5 diameter revolving at a speed of 800 2 rem, under a head of 20 m. What is the diameter and output of another turbine which works under a head of 180 m at speed of 200 RPM when their efficiencies are the same. Find the specific speed and name the turbdine that can be used

{15 Marks]